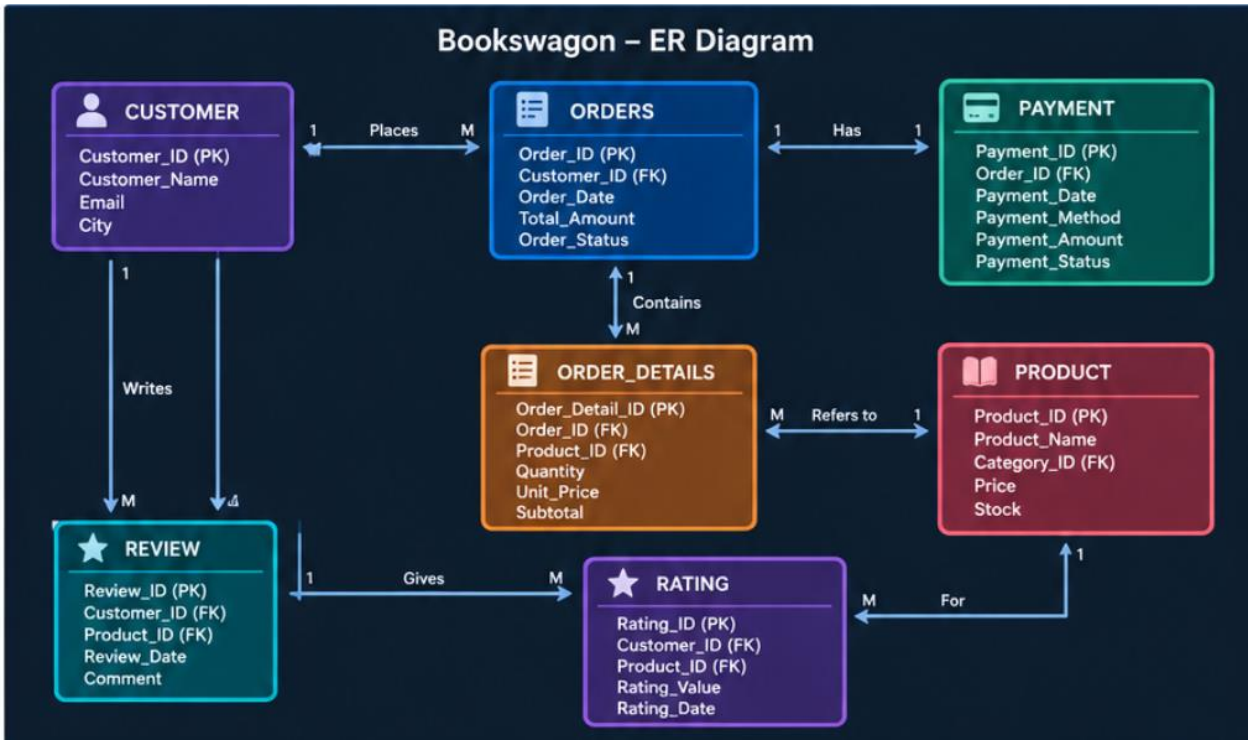
DBMS – DATABASE RELATIONSHIP ANALYSIS USING JOINS

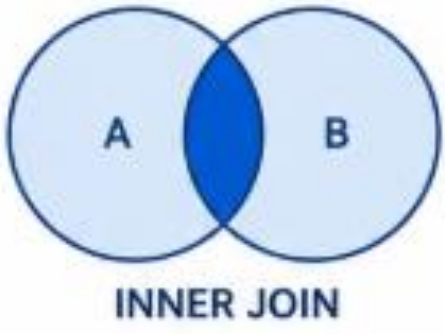
Project Title: Bookswagon – Online Bookstore Online Order Management System

WEEK 8

1.Combining Multiple Tables



2. INNER JOIN



2.1 Customer and Orders

```
SQL> SELECT c.Customer_ID,
  2         c.Customer_Name,
  3         o.Order_ID,
  4         o.Total_Amount
  5  FROM BookswagonCustomer c
  6  INNER JOIN BookswagonOrders o
  7    ON c.Customer_ID = o.Customer_ID;
```

```
CUSTOMER_ID CUSTOMER_NAME          ORDER_ID ORDER_DATE   TOTAL_AMOUNT
----------- -------------------- ---------- ----------- ------------
          1 Nandhini Sri               5001 18-SEP-26         1499.00
          2 Priya Kumar                5002 22-SEP-26          749.00
          3 Arun Kumar                 5003 19-SEP-26          899.00
          4 Divya Raj                  5004 19-SEP-26          775.00
          5 Karthik S                  5005 20-SEP-26          899.00
          6 Harini Devi                5006 20-SEP-26          749.00
          7 Rahul Kumar                5007 20-SEP-26         1197.00
          8 Keerthana R                5008 21-SEP-26          998.00
          9 Vignesh M                  5009 21-SEP-26          749.00
         10 Swetha Priya               5010 21-SEP-26          900.00
```

2. Orders and Payment — INNER JOIN

```
SQL> SELECT o.Order_ID,
  2         o.Order_Date,
  3         o.Total_Amount,
  4         p.Payment_Method,
  5         p.Payment_Amount,
  6         p.Payment_Status
  7  FROM BookswagonOrders o
  8  INNER JOIN BookswagonPayment p
  9  ON o.Order_ID = p.Order_ID;
```

```
 ORDER_ID ORDER_DATE   TOTAL_AMOUNT PAYMENT_METHOD                PAYMENT_AMOUNT PAYMENT_STATUS
---------- ------------ ------------ ------------------------------ -------------- ---------------
     5001 18-SEP-26         1499.00 UPI                                    1250.00 Paid
     5002 22-SEP-26          749.00 Credit Card                             898.00 Paid
     5003 19-SEP-26          899.00 Debit Card                             1196.00 Paid
     5004 19-SEP-26          775.00 UPI                                     775.00 Paid
     5005 20-SEP-26          899.00 Cash on Delivery                       1025.00 Paid
     5006 20-SEP-26          749.00 UPI                                     450.00 Paid
     5007 20-SEP-26         1197.00 Net Banking                             897.00 Paid
     5008 21-SEP-26          998.00 Credit Card                             998.00 Paid
     5009 21-SEP-26          749.00 UPI                                     550.00 Paid
     5010 21-SEP-26          900.00 Cash on Delivery                        700.00 Failed

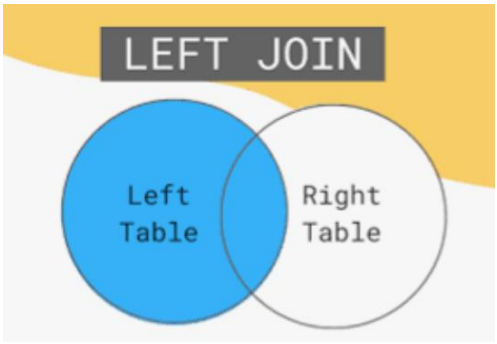
10 rows selected.
```

2.3 Product and Order_details

```
SQL> SELECT p.Product_ID,
  2         p.Product_Name,
  3         od.Order_ID,
  4         od.Quantity,
  5         od.Unit_Price,
  6         od.Subtotal
  7  FROM BookswagonProduct p
  8  INNER JOIN BookswagonOrderDetails od
  9  ON p.Product_ID = od.Product_ID;
```

```
PRODUCT_ID PRODUCT_NAME                    ORDER_ID   QUANTITY UNIT_PRICE  SUBTOTAL
---------- ------------------------------ ---------- ---------- ---------- ---------
       101 The Alchemist                        1001          1     450.00    450.00
       102 Atomic Habits                        1002          1     499.00    499.00
       103 The Psychology of Money              1003          1     350.00    350.00
       104 Wings of Fire                        1004          1     299.00    299.00
       106 Rich Dad Poor Dad                    1005          1     399.00    399.00
```

3. LEFT JOIN



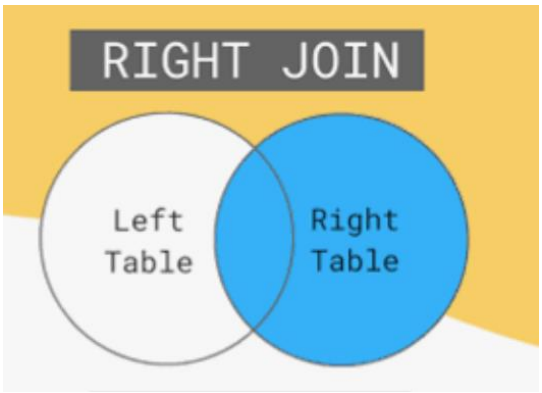
3.1 Display all Customers and Orders

```
SQL> SELECT c.Customer_ID,
  2         c.Customer_Name,
  3         o.Order_ID,
  4         o.Order_Date,
  5         o.Total_Amount
  6  FROM BookswagonCustomer c
  7  LEFT JOIN BookswagonOrders o
  8  ON c.Customer_ID = o.Customer_ID
  9  ORDER BY c.Customer_ID;
```

```
CUSTOMER_ID CUSTOMER_NAME          ORDER_ID ORDER_DATE   TOTAL_AMOUNT
----------- -------------------- ---------- ----------- ------------
          1 Nandhini Sri               1001 18-OCT-26         450.00
          1 Nandhini Sri               5001 18-SEP-26        1499.00
          2 Priya Kumar                5002 22-SEP-26         749.00
          2 Priya Kumar                1002 19-OCT-26         499.00
          3 Arun Kumar                 5003 19-SEP-26         899.00
          3 Arun Kumar                 1003 20-OCT-26         350.00
          4 Divya Raj                  5004 19-SEP-26         775.00
          4 Divya Raj                  1004 21-OCT-26         299.00
          5 Karthik S                  5005 20-SEP-26         899.00
          5 Karthik S                  1005 22-OCT-26         399.00
          6 Harini Devi                5006 20-SEP-26         749.00
          7 Rahul Kumar                5007 20-SEP-26        1197.00
          8 Keerthana R                5008 21-SEP-26         998.00
          9 Vignesh M                  5009 21-SEP-26         749.00
         10 Swetha Priya               5010 21-SEP-26         900.00
         11 Sanjay Kumar
         12 Aishwarya S
         13 Dinesh Raj
         14 Meena Lakshmi
         15 Adithya K

20 rows selected.
```

4. RIGHT JOIN



4.1. Display all orders and customers details

```
SQL> SELECT o.Order_ID,
  2         o.Order_Date,
  3         o.Total_Amount,
  4         o.Order_Status,
  5         c.Customer_ID,
  6         c.Customer_Name
  7  FROM BookswagonCustomer c
  8  RIGHT JOIN BookswagonOrders o
  9  ON c.Customer_ID = o.Customer_ID
 10  ORDER BY o.Order_ID;
```

ORDER_ID	ORDER_DATE	TOTAL_AMOUNT	ORDER_STATUS	CUSTOMER_ID	CUSTOMER_NAME
1001	18-OCT-26	450.00	Delivered	1	Nandhini Sri
1002	19-OCT-26	499.00	Shipped	2	Priya Kumar
1003	20-OCT-26	350.00	Pending	3	Arun Kumar
1004	21-OCT-26	299.00	Delivered	4	Divya Raj
1005	22-OCT-26	399.00	Pending	5	Karthik S
5001	18-SEP-26	1499.00	Delivered	1	Nandhini Sri
5002	22-SEP-26	749.00	Shipped	2	Priya Kumar
5003	19-SEP-26	899.00	Processing	3	Arun Kumar
5004	19-SEP-26	775.00	Delivered	4	Divya Raj
5005	20-SEP-26	899.00	Shipped	5	Karthik S
5006	20-SEP-26	749.00	Processing	6	Harini Devi
5007	20-SEP-26	1197.00	Delivered	7	Rahul Kumar
5008	21-SEP-26	998.00	Processing	8	Keerthana R
5009	21-SEP-26	749.00	Shipped	9	Vignesh M
5010	21-SEP-26	900.00	Processing	10	Swetha Priya

15 rows selected.

5. Complete Order Details

```
SQL> SELECT c.Customer_Name,
  2         o.Order_ID,
  3         TO_CHAR(o.Order_Date, 'DD-MON-YY') AS Order_Date,
  4         p.Product_ID,
  5         p.Product_Name,
  6         od.Quantity,
  7         od.Unit_Price,
  8         od.Subtotal,
  9         py.Payment_Method,
 10         py.Payment_Amount,
 11         py.Payment_Status
 12  FROM BookswagonCustomer c
 13  JOIN BookswagonOrders o
 14    ON c.Customer_ID = o.Customer_ID
 15  JOIN BookswagonOrderDetails od
 16    ON o.Order_ID = od.Order_ID
 17  JOIN BookswagonProduct p
 18    ON od.Product_ID = p.Product_ID
 19  JOIN BookswagonPayment py
 20    ON o.Order_ID = py.Order_ID
 21  ORDER BY o.Order_Date DESC;
```

```
CUSTOMER_NAME   ORDER_ID ORDER_DATE PRODUCT_ID PRODUCT_NAME             QUANTITY UNIT_PRICE  SUBTOTAL PAYMENT_METHOD  PAYMENT_AMOUNT PAYMENT_STATUS
--------------- -------- ---------- ---------- ------------------------ -------- ---------- --------- --------------- -------------- --------------
Nandhini Sri        5001 18-SEP-26         101 The Alchemist                   1     450.00    450.00 UPI                    1250.00 Paid
```

6. Customer Purchase History

```
SQL> SELECT c.Customer_ID,
  2         c.Customer_Name,
  3         o.Order_ID,
  4         TO_CHAR(o.Order_Date, 'DD-MON-YYYY') AS Order_Date,
  5         p.Product_Name,
  6         od.Quantity,
  7         od.Subtotal
  8  FROM BookswagonCustomer c
  9  INNER JOIN BookswagonOrders o
 10      ON c.Customer_ID = o.Customer_ID
 11  INNER JOIN BookswagonOrderDetails od
 12      ON o.Order_ID = od.Order_ID
 13  INNER JOIN BookswagonProduct p
 14      ON od.Product_ID = p.Product_ID
 15  ORDER BY c.Customer_ID, o.Order_Date;
```

```
CUSTOMER_ID CUSTOMER_NAME        ORDER_ID ORDER_DATE   PRODUCT_NAME                    QUANTITY  SUBTOTAL
----------- -------------------- -------- ------------ ------------------------------ --------- ---------
          1 Priya                   5001 18-SEP-2026  The Alchemist                          1    450.00
          1 Priya                   1001 18-OCT-2026  The Alchemist                          1    450.00
          2 Priya Kumar             1002 19-OCT-2026  Atomic Habits                          1    499.00
          2 Priya Kumar             1002 19-OCT-2026  The Alchemist                          1    450.00
          3 Arun Kumar              1003 20-OCT-2026  The Psychology of Money                1    350.00
          3 Arun Kumar              1003 20-OCT-2026  The Alchemist                          1    450.00
          4 Divya Raj               1004 21-OCT-2026  Wings of Fire                          1    299.00
          4 Divya Raj               1004 21-OCT-2026  The Alchemist                          1    450.00
          5 Karthik S               1005 22-OCT-2026  The Alchemist                          1    450.00
          5 Karthik S               1005 22-OCT-2026  Rich Dad Poor Dad                      1    399.00

10 rows selected.
```

7. Customer Purchase history with payment

```
SQL> SELECT c.Customer_ID,
  2         c.Customer_Name,
  3         o.Order_ID,
  4         TO_CHAR(o.Order_Date, 'DD-MON-YYYY') AS Order_Date,
  5         p.Product_Name,
  6         od.Quantity,
  7         od.Subtotal,
  8         py.Payment_Method,
  9         py.Payment_Status
 10  FROM BookswagonCustomer c
 11  INNER JOIN BookswagonOrders o
 12      ON c.Customer_ID = o.Customer_ID
 13  INNER JOIN BookswagonOrderDetails od
 14      ON o.Order_ID = od.Order_ID
 15  INNER JOIN BookswagonProduct p
 16      ON od.Product_ID = p.Product_ID
 17  INNER JOIN BookswagonPayment py
 18      ON o.Order_ID = py.Order_ID
 19  ORDER BY c.Customer_ID;
```

```
CUSTOMER_ID CUSTOMER_NAME      ORDER_ID ORDER_DATE   PRODUCT_NAME                QUANTITY  SUBTOTAL PAYMENT_METHOD  PAYMENT_STATUS
----------- ------------------ -------- ------------ ---------------------------- --------- --------- --------------- ---------------
          1 Priya                 5001 18-SEP-2026  The Alchemist                       1    450.00 UPI             Paid
```

8. Multi-Table Business Report

```
SQL> SELECT c.Customer_Name,
  2         COUNT(DISTINCT o.Order_ID) AS Total_Orders,
  3         SUM(od.Subtotal) AS Total_Purchase,
  4         SUM(od.Quantity) AS Total_Products
  5  FROM BookswagonCustomer c
  6  INNER JOIN BookswagonOrders o
  7      ON c.Customer_ID = o.Customer_ID
  8  INNER JOIN BookswagonOrderDetails od
  9      ON o.Order_ID = od.Order_ID
 10  GROUP BY c.Customer_Name
 11  ORDER BY Total_Purchase DESC;
```

```
CUSTOMER_NAME        TOTAL_ORDERS TOTAL_PURCHASE TOTAL_PRODUCTS
-------------------- ------------ -------------- --------------
Priya Kumar                     1         949.00              2
Priya                           2         900.00              2
Karthik S                       1         849.00              2
Arun Kumar                      1         800.00              2
Divya Raj                       1         749.00              2
```

9. Payment Transaction Report

```
SQL> SELECT c.Customer_Name,
  2         o.Order_ID,
  3         TO_CHAR(o.Order_Date, 'DD-MON-YY') AS Order_Date,
  4         o.Total_Amount,
  5         py.Payment_Method,
  6         py.Payment_Amount,
  7         py.Payment_Status
  8  FROM BookswagonCustomer c
  9  INNER JOIN BookswagonOrders o
 10      ON c.Customer_ID = o.Customer_ID
 11  INNER JOIN BookswagonPayment py
 12      ON o.Order_ID = py.Order_ID
 13  ORDER BY o.Order_Date DESC;
```

CUSTOMER_NAME	ORDER_ID	ORDER_DATE	TOTAL_AMOUNT	PAYMENT_METHOD	PAYMENT_AMOUNT	PAYMENT_STATUS
Priya Kumar	5002	22-SEP-26	749.00	Credit Card	898.00	Paid
Vignesh M	5009	21-SEP-26	749.00	UPI	550.00	Paid
Keerthana R	5008	21-SEP-26	998.00	Credit Card	998.00	Paid
Swetha Priya	5010	21-SEP-26	900.00	Cash on Deliver	700.00	Failed
Karthik S	5005	20-SEP-26	899.00	Cash on Deliver	1025.00	Paid
Rahul Kumar	5007	20-SEP-26	1197.00	Net Banking	897.00	Paid
Harini Devi	5006	20-SEP-26	749.00	UPI	450.00	Paid
Arun Kumar	5003	19-SEP-26	899.00	Debit Card	1196.00	Paid
Divya Raj	5004	19-SEP-26	775.00	UPI	775.00	Paid
Priya	5001	18-SEP-26	1499.00	UPI	1250.00	Paid

10 rows selected.

